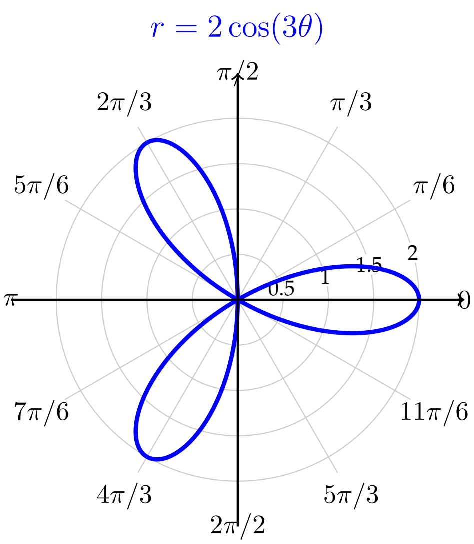
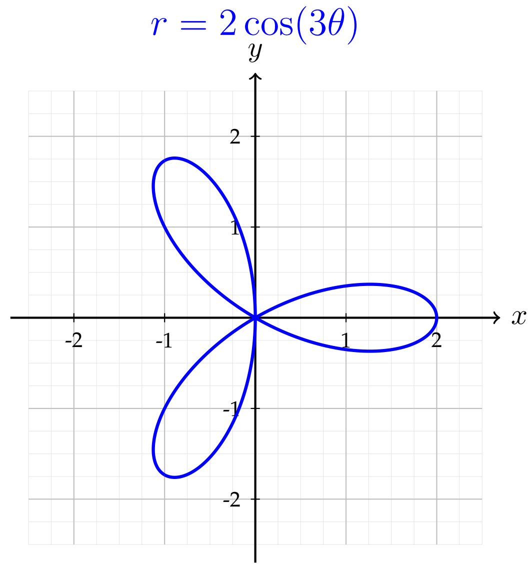
Section 7.4: Slope and Arc Length in Polar Coordinates (Extra)

Recall the polar curve $r = f(\theta) = 2\cos(3\theta)$, graphed below.





(1) On the graphs above, sketch a tangent line at $\theta = \pi/6$ and another one at $\theta = \pi/3$. Next to your tangent lines, estimate the slope.

(2) Find $f'(\theta)$ and evaluate it at $\theta = \pi/6$ and $\theta = \pi/3$. Does this fit with your estimates above?

(3) Review for the Final Exam
If we have a parametrized curve $x(t)$, $y(t)$, how do we find dy/dx?

(4) Use this to find dy/dx for the polar curve and evaluate this at $\theta = \pi/3$.

(5) Review for the Final Exam
If we have a parametrized curve $x(t)$, $y(t)$, how do we find its arc length from $t = \alpha$ to $t = \beta$?

(6) Use this formula to set up an integral to find the arc length in one petal of the 3-petal flower $r = f(\theta) = 2\cos(3\theta)$.